

CTD Action Taxonomy Report

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Total CTD edges
1109476

STRICT edges
1028760

Pruned edges
80716

Pruned %
7.28%

STRICT = edges after removing generic “affects^expression / response to substance / multi-affects chains”. Families kept: activates | inhibits | binds | modifies. Focus on binds/modifies for mechanistic leads.

Family distribution (STRICT only)

Dataset shape (FULL vs STRICT)

Edges (FULL)	1109476
Edges (STRICT)	1028760
Edges pruned	80716
Pairs (FULL)	841342
Pairs (STRICT)	789969
Pairs pruned	51373

DDI badges (top 150)

Badge logic combines STRICT “activates/inhibits” and FULL ↑ / ↓ expression on key ADME genes (CYPs, ABCB1, SLC01B1, UGTs).\ Check DDI: ↑ / ↓ /mixed indicates potential induction/inhibition risk worth ADME/Tox review.

chem_id	chem_name	induces_n	inhibits_n	induces_
MESH:C076994	perfluorooctane sulfonic acid	5	10	CYP2C1
MESH:D011794	Quercetin	7	10	ABCB1 C
MESH:D012293	Rifampin	10	5	ABCB1 C
MESH:D000082	Acetaminophen	6	9	ABCB1 C
MESH:D016604	Aflatoxin B1	8	9	ABCB1 C
MESH:C006780	bisphenol A	9	5	ABCB1 C
MESH:D002635	Chenodeoxycholic Acid	4	9	CYP1A2
MESH:D016572	Cyclosporine	3	9	ABCB1 C
MESH:D007654	Ketoconazole	7	9	ABCB1 C

chem_id	chem_name	induces_n	inhibits_n	induces_
MESH:D019319	Okadaic Acid	3	9	ABCB1 C
MESH:C017947	sodium arsenite	4	9	ABCB1 C
MESH:D014260	Triclosan	9	1	ABCB1 C
MESH:D014635	Valproic Acid	9	9	ABCB1 C

Top-3 Reactome pathways per chemical (top 150)

Top-3 Reactome pathways per compound using STRICT edges (distinct genes per pathway).

chem_id	chem_name	top3	
MESH:C534883	10074-G5	Signal Transduction (2); Generic Transcription Pathway (2); Cellular responses to stress (2)	
MESH:C112297	"10	10-bis(4-pyridinylmethyl)-9(10H)-anthracenone"	N P V c
MESH:C004822	"10	11-dihydro-10	1: di
MESH:C039775	"10	11-dihydro-10-hydroxycarbamazepine"	D (1
MESH:C496879	103D5R	Cytokine Signaling in Immune system (3); Signal Transduction (3); Cell Cycle (3)	
MESH:C553100	10-(4'-(N-diethylamino)butyl)-2-chlorophenoxazine	"Hemostasis (4); Platelet activation	si PI Fo
MESH:C532038	10-(6'-plastoquinonyl)decyltriphenylphosphonium	Disease (4); Metabolism of proteins (4); Post-translational protein	

Counts after STRICT pruning.

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inhibits	510037
activates	485690
modifies	25154
binds	7879

Full vs STRICT by gene (only genes with changes)

Flags high-prune genes ($\geq 25\%$ pruned) and highlights presence of binds/modifies in STRICT.

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TP53	other	147	0	147	high_prune	mechanistic_signal	High prune: likely expression-only; rely on STRICT. STRICT has binds/modifies → good mechanistic lead.
BAX	other	120	0	120	high_prune	mechanistic_signal	High prune: likely expression-only; rely on STRICT. STRICT has binds/modifies → good mechanistic lead.
BCL2	other	119	0	119	high_prune	mechanistic_signal	High prune: likely expression-only; rely on STRICT.

Top chemicals by pruned edges

High pruned counts often reflect expression-only breadth; deprioritize for causal inference.

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MESH:C005445	triphenyl phosphate	7409
MESH:D014635	Valproic Acid	6718
MESH:C065087	di-n-butylphosphoric acid	5761
MESH:D008070	Lipopolysaccharides	2531
MESH:D014028	Tobacco Smoke Pollution	2327
MESH:C006780	bisphenol A	2298
MESH:D000393	Air Pollutants	2131
MESH:D010126	Ozone	2067

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MESH:C039961	S-(1,2-dichlorovinyl)cysteine	1999
MESH:D006861	Hydrogen Peroxide	1995
MESH:D002220	Carbamazepine	1984
MESH:C044887	beta-methylcholine	1699
MESH:D007854	Lead	1614
MESH:D016604	Aflatoxin B1	1609

Sample of pruned edges (names + patterns)

Concrete examples of what STRICT removes, useful for QA.

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A1BG	MESH:C033674	benazol P	affects^expression
A1BG	MESH:D008628	Mercury	affects^expression
A1BG	MESH:D013749	Tetrachlorodibenzodioxin	affects^expression
A1BG	MESH:D014635	Valproic Acid	affects^expression

Methods: public.ctd_action_family; STRICT excludes generic affects^expression/response chains. Sources: CTD chemicals & chem-gene edges already ingested.